Spectrum
SPC

From Binary to Beyond

بنك أسئلة فيزياء عامة 1

الوحدات 1–3 · 150 سؤال اختيار من متعدد

الوحدة 1: Physics and Measurement

أول 15 سؤالاً سهلة، ثم 20 متوسطة، وآخر 15 صعبة.

ثوابت للاستخدام: 1 in = 2.54 cm، 1 mi = 1609 m، السنة = 3.156×10^{7} s، $c = 3.00\times10^{8}$ m/s.

1 Which of these is an SI base unit?

A newton
B joule
C kilogram
D watt

2 How many significant figures are in 0.00450 m?

A 3
B 5
C 6
D 2

3 5.0 km equals

A 50 m
B 500 m
C 0.50 m
D 5.0×10^{3} m

4 The SI prefix for 10^{-3} is

A micro
B milli
C centi
D kilo

5 2.5 h equals

A 9.0×10^{3} s
B 1.5×10^{2} s
C 2.5×10^{3} s
D 3.6×10^{3} s

6 3.2×10^{-6} m can be written as

A 3.2 mm
B 3.2 nm
C 3.2 pm
D 3.2 µm

7 45,000,000 in scientific notation is

A 4.5×10^{6}
B 4.5×10^{8}
C 4.5×10^{7}
D 4.5×10^{-7}

8 How many significant figures are in 1.0050?

A 4
B 5
C 3
D 6

9 3.14159 rounded to three significant figures is

A 3.1
B 3.15
C 3.142
D 3.14

10 1.00 in equals

A 2.54 cm
B 25.4 cm
C 0.254 cm
D 10.0 cm

11 The dimensions of velocity are

A $L \cdot T$
B L/T
C L/T^2
D T/L

12 The dimensions of acceleration are

A L/T
B $L \cdot T^2$
C L/T^2
D T^2/L

13 The order of magnitude of 3,200 is

A 10^3
B 10^4
C 10^2
D 10^5

14 72 km/h equals

A 7.2 m/s
B 2.0 m/s
C 260 m/s
D 20 m/s

15 12.3 m + 4.56 m, with correct significant figures, is

A 16.86 m
B 16.9 m
C 17 m
D 16.8 m

16 2.0 m × 3.45 m, with correct significant figures, is

A $6.90\ m^2$
B $7\ m^2$
C $6.9\ m^2$
D $6.9\times10^{1}\ m^2$

17 $(3.0\times10^{4})(2.0\times10^{-2})$ =

A 6.0×10^{2}
B 6.0×10^{-8}
C 6.0×10^{6}
D 5.0×10^{2}

18 $100\ cm^2$ equals

A $1.0\ m^2$
B $0.10\ m^2$
C $1.0\times10^{-4}\ m^2$
D $1.00\times10^{-2}\ m^2$

19 $1.00\ m^3$ equals

A $10^{3}\ cm^3$
B $10^{6}\ cm^3$
C $10^{4}\ cm^3$
D $10^{9}\ cm^3$

20 The dimensions of force are

A $M\cdot L\cdot T^{-1}$
B $M\cdot L^{2}\cdot T^{-2}$
C $M\cdot L\cdot T^{-2}$
D $M\cdot L^{-1}\cdot T^{-2}$

21 Is $v = at$ dimensionally consistent?

A Yes
B No
C Only if $t = 1\ s$
D It cannot be determined

22 Is $x = (1/2)\,a\,t^2$ dimensionally consistent?

A No
B Yes
C Only if $a = 0$
D It cannot be determined

23 Is $x = v\,t^2$ dimensionally consistent?

A Yes
B Yes, if v is constant
C It cannot be determined
D No, the right side has dimensions $L\cdot T$

24 The volume of a cube of side 2.00 cm is

A $8.00\times10^{-4}\ m^3$
B $8.00\times10^{-2}\ m^3$
C $8.00\times10^{-6}\ m^3$
D $4.00\times10^{-6}\ m^3$

25 A block has mass 500 g and volume 250 cm^3. Its density is

A $2.0\ g/cm^3$ ($2.0\times10^3\ kg/m^3$)
B $0.50\ g/cm^3$
C $2.0\times10^6\ kg/m^3$
D $125\ g/cm^3$

26 A density of 1.0 g/cm^3 equals

A $1.0\ kg/m^3$
B $10\ kg/m^3$
C $100\ kg/m^3$
D $1.0\times10^3\ kg/m^3$

27 The mass of 3.0 L of water is

A 0.30 kg
B 3.0 kg
C 30 kg
D 3.0 g

28 In F = kx, the constant k has dimensions

A $M\cdot T^{-2}$
B $M\cdot L\cdot T^{-2}$
C $M\cdot L^2\cdot T^{-2}$
D $M\cdot T^2$

29 The dimensions of power are

A $M\cdot L\cdot T^{-2}$
B $M\cdot L^2\cdot T^{-2}$
C $M\cdot L^2\cdot T^{-3}$
D $M\cdot L\cdot T^{-3}$

30 60 mi/h (1 mi = 1609 m) equals

A 96.5 m/s
B 16.7 m/s
C 60 m/s
D 26.8 m/s

31 The area of a circle of radius 3.50 cm is

A $3.85\times10^{-2}\ m^2$
B $3.85\times10^{-3}\ m^2$
C $1.10\times10^{-3}\ m^2$
D $3.85\times10^{-4}\ m^2$

32 The volume of a sphere of radius 0.10 m is

A $4.19\times10^{-2}\ m^3$
B $1.26\times10^{-2}\ m^3$
C $4.19\times10^{-3}\ m^3$
D $3.14\times10^{-2}\ m^3$

33 A plate is 21.3 cm by 9.8 cm. Its area with correct significant figures is

A $2.1\times10^{2}\ cm^2$
B $208.74\ cm^2$
C $2.09\times10^{2}\ cm^2$
D $2.087\times10^{2}\ cm^2$

34 The mass of air (1.2 kg/m^3) in a 5.0 m × 4.0 m × 3.0 m room is about

A 7.2 kg
B 1.2 kg
C 720 kg
D 72 kg

35 Gold has density 19.3 g/cm^3. A gold cube of side 2.00 cm has mass

A 38.6 g
B 154 g
C 77 g
D 19.3 g

36 Light (3.00×10^{8} m/s) travels how far in 1.0 ns?

A 30 cm
B 3.0 m
C 3.0 cm
D 0.30 cm

37 The number of seconds in one year is about

A 3.16×10^{5}
B 3.16×10^{6}
C 3.16×10^{7}
D 3.16×10^{9}

38 Is $E = mc^2$ dimensionally consistent?

A No
B Yes, both sides have dimensions $M\cdot L^2\cdot T^{-2}$
C Only for light
D Yes, both sides have dimensions $M\cdot L\cdot T^{-2}$

39 If $K = (1/2)\,m\,v^n$ has the dimensions of energy, then n =

A 1
B 3
C 1/2
D 2

40 How many atoms of diameter 1.0×10^{-10} m placed side by side span 1.0 mm?

A 1.0×10^{7}
B 1.0×10^{13}
C 1.0×10^{-7}
D 1.0×10^{10}

41 A solid sphere has mass 0.50 kg and radius 5.0 cm. Its average density is about

A 9.5×10^{4} kg/m^3
B 2.4×10^{3} kg/m^3
C 9.5×10^{2} kg/m^3
D 1.0×10^{3} kg/m^3

42 A cylindrical tank of radius 1.5 m and height 2.4 m holds about

A 1.7×10^{3} L
B 1.7×10^{4} L
C 1.7×10^{2} L
D 5.4×10^{4} L

43 15 m/s^2 equals

A 54 km/h^2
B 1.5×10^{-2} km/h^2
C 5.4×10^{4} km/h^2
D 1.9×10^{5} km/h^2

44 In F = G m1 m2/r^2, the dimensions of G are

A $L^3\cdot M^{-1}\cdot T^{-2}$
B $L^2\cdot M^{-1}\cdot T^{-2}$
C $L^3\cdot M\cdot T^{-2}$
D $M\cdot L^{-3}\cdot T^2$

45 For a pendulum, $T = C\,L^a\,g^b$. Dimensional analysis gives

A $a = -1/2,\ b = 1/2$
B $a = 1,\ b = -1$
C $a = 1/2,\ b = -1/2$
D $a = 2,\ b = -2$

46 Centripetal acceleration $a = k\,v^p\,r^q$ (k dimensionless). Then

A $p = 1,\ q = -1$
B $p = 2,\ q = -1$
C $p = 2,\ q = 1$
D $p = 1,\ q = -2$

47 The side of a 1.00 kg aluminum cube (density 2700 kg/m^3) is about

A 0.37 m
B 7.18 m
C 0.718 m
D 7.18 cm

48 A rectangle measures 12.0 ± 0.2 cm by 5.0 ± 0.1 cm. Its area is

A $60 \pm 2\ \text{cm}^2$
B $60 \pm 0.3\ \text{cm}^2$
C $60 \pm 0.02\ \text{cm}^2$
D $60 \pm 6\ \text{cm}^2$

49 One light-year is about

A $9.47\times10^{12}\ \text{m}$
B $9.47\times10^{18}\ \text{m}$
C $9.47\times10^{15}\ \text{m}$
D $3.00\times10^{8}\ \text{m}$

50 The number of heartbeats in 70 years at 70 beats/min is about

A 2.6×10^{7}
B 2.6×10^{9}
C 2.6×10^{11}
D 2.6×10^{6}

الوحدة 2: Motion in One Dimension

أول 15 سؤالاً سهلة، ثم 20 متوسطة، وآخر 15 صعبة.

استخدم $g = 9.80\ m/s^2$ وأهمل مقاومة الهواء.

51 A runner goes 100 m east then 40 m west. The distance traveled and displacement are

A 60 m; 140 m
B 140 m; 60 m east
C 140 m; 140 m east
D 60 m; 60 m

52 A car travels 120 km in 2.0 h. Its average speed is

A 240 km/h
B 0.017 km/h
C 120 km/h
D 60 km/h

53 An object moves from x = 2.0 m to x = 10.0 m in 4.0 s. Its average velocity is

A 2.0 m/s
B 3.0 m/s
C 0.5 m/s
D 8.0 m/s

54 The SI unit of acceleration is

A m/s
B m^2/s
C m/s^2
D N

55 A car moves at a constant 20 m/s for 5.0 s. Its displacement is

A 25 m
B 100 m
C 4.0 m
D 10 m

56 The average velocity of an object that returns to its starting point is

A zero
B twice its average speed
C half its average speed
D equal to its average speed

57 A car accelerates uniformly from rest to 30 m/s in 6.0 s. Its acceleration is

A $0.20\ m/s^2$
B $180\ m/s^2$
C $24\ m/s^2$
D $5.0\ m/s^2$

58 A particle has $v = 4 + 3t$ (SI units). Then

A $v_0 = 3$ m/s, $a = 4$ m/s^2
B $v_0 = 4$ m/s, $a = 4$ m/s^2
C $v_0 = 4$ m/s, $a = 3$ m/s^2
D $v_0 = 7$ m/s, $a = 3$ m/s^2

59 A ball is dropped from rest. Its speed after 2.0 s is

A 19.6 m/s
B 9.8 m/s
C 4.9 m/s
D 39.2 m/s

60 A ball is dropped from rest. It falls how far in 3.0 s?

A 29.4 m
B 44.1 m
C 88.2 m
D 14.7 m

61 At the top of its path, a projectile's acceleration (ignoring air resistance) is

A zero
B 9.80 m/s^2 upward
C dependent on its speed
D 9.80 m/s^2 downward

62 A car slows from 25 m/s to 15 m/s in 4.0 s. Its acceleration is

A +2.5 m/s^2
B -10 m/s^2
C -2.5 m/s^2
D -6.25 m/s^2

63 A particle has $x = 5 + 2t$ (SI units). Its velocity is

A 5 m/s
B 2 m/s, constant
C 7 m/s
D 2t m/s

64 A car starts from rest with $a = 2.0$ m/s^2 for 10 s. Its final speed and distance are

A 20 m/s; 100 m
B 20 m/s; 200 m
C 10 m/s; 50 m
D 5 m/s; 25 m

65 Which statement is correct?

A The slope of an x-t graph is acceleration.
B The slope of a v-t graph is velocity.
C The slope of x-t is velocity; the slope of v-t is acceleration; the area under v-t is displacement.
D The area under a v-t graph is acceleration.

66 A particle has $v = 10 - 2t$ (SI units). It stops momentarily at

A $t = 2$ s
B $t = 10$ s
C $t = 20$ s
D $t = 5.0$ s

67 A car drives 1.0 h at 60 km/h then 1.0 h at 40 km/h. Its average speed is

A 40 km/h
B 50 km/h
C 48 km/h
D 100 km/h

68 A car drives the first half of the distance at 60 km/h and the second half at 40 km/h. Its average speed is

A 48 km/h
B 50 km/h
C 100 km/h
D 24 km/h

69 An object at 10 m/s decelerates at 2.0 m/s^2 until it stops. It travels

A 5 m
B 50 m
C 25 m
D 10 m

70 A car brakes from 20 m/s at 4.0 m/s^2. The stopping time and distance are

A 5.0 s; 100 m
B 10 s; 100 m
C 5.0 s; 25 m
D 5.0 s; 50 m

71 A stone is thrown up at 14.7 m/s. The time to the top and the maximum height are

A 1.5 s; 11.0 m
B 3.0 s; 22.0 m
C 1.5 s; 22.0 m
D 3.0 s; 11.0 m

72 A particle has $x = 3t^2 - 2t + 1$ (SI units). At $t = 2.0$ s,

A $v = 12$ m/s, $a = 6$ m/s^2
B $v = 10$ m/s, $a = 6$ m/s^2
C $v = 10$ m/s, $a = 12$ m/s^2
D $v = 6$ m/s, $a = 10$ m/s^2

73 A particle has $x = 2t^3 - 5t^2 + 3$ (SI units). At $t = 3.0$ s,

A $v = 24$ m/s, $a = 12$ m/s^2
B $v = 54$ m/s, $a = 26$ m/s^2
C $v = 24$ m/s, $a = 18$ m/s^2
D $v = 24$ m/s, $a = 26$ m/s^2

74 A train from rest accelerates at 0.50 m/s^2. To reach 90 km/h (25 m/s) it needs

A 25 s and 312 m
B 50 s and 1250 m
C 50 s and 625 m
D 180 s and 625 m

75 A ball is dropped from an 80 m building. The fall time and impact speed are

A 4.04 s; 39.6 m/s
B 8.16 s; 80 m/s
C 4.04 s; 19.8 m/s
D 2.02 s; 39.6 m/s

76 A ball is thrown downward at 5.0 m/s from 30 m. Its speed at the ground is

A 17.1 m/s
B 24.8 m/s
C 24.3 m/s
D 35.0 m/s

77 A ball is thrown up at 20 m/s from the ground. It is 15 m high at

A 0.5 s and 3.5 s
B 1.0 s only
C 0.99 s and 3.09 s
D 1.5 s and 2.5 s

78 A car with v0 = 15 m/s accelerates at 3.0 m/s^2 over 100 m. Its final speed and time are

A 22.4 m/s; 2.5 s
B 28.7 m/s; 9.6 s
C 25 m/s; 3.3 s
D 28.7 m/s; 4.57 s

79 A v-t graph rises linearly from 0 to 20 m/s in 5.0 s, stays at 20 m/s for 10 s, then falls linearly to 0 in 5.0 s. The total displacement is

A 200 m
B 300 m
C 250 m
D 400 m

80 A rock falls past a 2.0 m window in 0.20 s. Its speed at the top of the window is

A 9.02 m/s
B 10.0 m/s
C 10.2 m/s
D 11.0 m/s

81 A car decelerates uniformly from 30 m/s to rest in 90 m. Its acceleration and stopping time are

A -5.0 m/s^2; 3.0 s
B -10 m/s^2; 6.0 s
C -5.0 m/s^2; 12 s
D -5.0 m/s^2; 6.0 s

82 A driver at 25 m/s has a 0.50 s reaction time, then brakes at 6.0 m/s^2. The total stopping distance is

A 52.1 m
B 77.1 m
C 64.6 m
D 12.5 m

83 A particle starts from rest at the origin with a = 6t (SI units). At t = 2.0 s,

A v = 12 m/s, x = 8 m
B v = 12 m/s, x = 24 m
C v = 6 m/s, x = 8 m
D v = 24 m/s, x = 16 m

84 A ball is thrown up at 10 m/s from a 25 m roof. The time to reach the ground and the impact speed are

A 3.50 s; 10.0 m/s
B 3.50 s; 24.3 m/s
C 2.04 s; 20.0 m/s
D 5.10 s; 40.0 m/s

85 Car A (x = 0) moves at 20 m/s; car B (x = 300 m) moves toward it at 30 m/s. They meet at

A 10 s; x = 200 m
B 15 s; x = 300 m
C 6.0 s; x = 200 m
D 6.0 s; x = 120 m

86 An elevator accelerates up from rest at 2.0 m/s^2 for 3.0 s, moves at constant speed for 8.0 s, then decelerates at 3.0 m/s^2 to rest. The total height is

A 48 m
B 57 m
C 63 m
D 72 m

87 A particle has v = 3t^2 - 12 (SI units). It is at rest at t = ___ and its acceleration there is ___

A 4 s; 24 m/s^2
B 2.0 s; 12 m/s^2
C 2.0 s; 6 m/s^2
D 12 s; 72 m/s^2

88 For v = 3t^2 - 12 with x(0) = 0, from t = 0 to 3.0 s the displacement and total distance are

A -9 m; 23 m
B -9 m; 9 m
C 9 m; 23 m
D -23 m; 23 m

89 A particle has $x = t^3/3 - 3t^2 + 8t$ (SI units). Its acceleration is zero at

A t = 3 s, where v = +1 m/s and x = 6 m
B t = 6 s, where v = 8 m/s
C t = 3.0 s, where v = -1 m/s and x = 6 m
D t = 2 s, where v = 0

90 A particle has a = 6t - 4, v(0) = 2.0 m/s, x(0) = 5.0 m. At t = 2.0 s,

A v = 6 m/s, x = 13 m
B v = 8 m/s, x = 9 m
C v = 6 m/s, x = 5 m
D v = 6 m/s, x = 9 m

91 A particle has v = 2t - 8 (SI units). From t = 0 to 6.0 s, the displacement and distance are

A -12 m; 20 m
B -12 m; 12 m
C 12 m; 20 m
D -20 m; 20 m

92 A ball is dropped from 40 m while another is thrown up from the ground at 20 m/s at the same time. They meet at

A 1.0 s; 30 m
B 2.0 s; 40 m
C 2.0 s; 20.4 m
D 4.0 s; 0 m

93 Car A starts from rest at x = 0 with $a = 4.0\ m/s^2$; car B starts from rest at x = 50 m with $a = 3.0\ m/s^2$. A catches B at

A 5 s; 50 m
B 10 s; 200 m
C 10 s; 250 m
D 20 s; 800 m

94 A car at 25 m/s passes a stationary car that starts from rest at $5.0\ m/s^2$. The second car catches up at

A 5 s; 125 m
B 10 s; 500 m
C 5 s; 250 m
D 10 s; 250 m

95 A train at 30 m/s sees another at 20 m/s, 400 m ahead on the same track. The minimum deceleration to avoid a collision is

A $0.25\ m/s^2$
B $0.125\ m/s^2$
C $1.0\ m/s^2$
D $0.50\ m/s^2$

96 A rocket accelerates up from rest at 20 m/s^2 for 10 s, then its engine stops. Its maximum height is about

A 3.04×10^3 m
B 1.0×10^3 m
C 2.0×10^3 m
D 4.1×10^3 m

97 A stone dropped into a well produces a splash heard 4.00 s later (sound speed 343 m/s). The well depth is about

A 78.4 m
B 59 m
C 70.5 m
D 1.37×10^3 m

98 A ball dropped from rest falls half its total height in the last 1.00 s. The fall time and height are

A 1.41 s; 9.8 m
B 2.00 s; 19.6 m
C 4.00 s; 78.4 m
D 3.41 s; 57.1 m

99 A particle has a = -0.50 v and v(0) = 10 m/s (SI units). Then v(4.0 s) and the total distance before it stops are

A 1.35 m/s; 20 m
B 1.35 m/s; 10 m
C 0 m/s; 10 m
D 5.0 m/s; 20 m

100 A particle moves with v = 2 sqrt(x) (SI units, x > 0). Its acceleration is

A 1.0 m/s^2
B constant, 2.0 m/s^2
C 4.0 m/s^2
D variable, proportional to 1/sqrt(x)

الوحدة 3: Vectors

أول 15 سؤالاً سهلة، ثم 20 متوسطة، وآخر 15 صعبة.

الزوايا تُقاس عكس عقارب الساعة من محور +x ما لم يُذكر غير ذلك.

101 Which list contains only scalar quantities?

A velocity, force
B displacement, mass
C speed, mass, temperature
D force, temperature

102 The point (3, 4) has polar coordinates

A $r = 5, \theta = 53.1°$
B $r = 7, \theta = 53.1°$
C $r = 5, \theta = 36.9°$
D $r = 25, \theta = 53.1°$

103 A vector of magnitude 10 along the +x axis has components

A (0, 10)
B (5, 5)
C (-10, 0)
D (10, 0)

104 The magnitude of $A = 3i + 4j$ is

A 7
B 5
C 25
D 12

105 If $A = 3i + 4j$ and $B = i - 2j$, then $A + B =$

A $2i + 6j$
B $4i + 6j$
C $4i + 2j$
D $3i + 2j$

106 For the same A and B, $A - B =$

A $2i + 6j$
B $4i + 2j$
C $2i + 2j$
D $-2i - 6j$

107 For $A = 3i + 4j$, $2A =$

A $6i + 4j$
B $5i + 6j$
C $3i + 8j$
D $6i + 8j$

108 The unit vector along A = 3i + 4j is

A 3i + 4j
B 0.6i + 0.8j
C 0.8i + 0.6j
D i + j

109 A person walks 3.0 m east then 4.0 m north. The displacement is

A 5.0 m at 53° north of east
B 7.0 m at 53° north of east
C 5.0 m at 37° north of east
D 5.0 m at 53° east of north

110 The components of a vector of magnitude 20 at 30° above the +x axis are

A (10.0, 17.3)
B (20, 10)
C (17.3, 10.0)
D (10, 10)

111 The polar coordinates r = 6.0, θ = 60° correspond to Cartesian coordinates

A (5.20, 3.0)
B (3.0, 5.20)
C (6, 60)
D (3.0, 3.0)

112 The point (-3, 3) has polar coordinates

A r = 4.24, θ = 45°
B r = 6, θ = 135°
C r = 4.24, θ = -45°
D r = 4.24, θ = 135°

113 Vector A is 5 units east and vector B is 5 units west. A + B is

A the zero vector
B 10 units east
C 10 units west
D 5 units east

114 The vector 6i - 8j has

A magnitude 14, direction 53° above +x
B magnitude 2, direction 53° below +x
C magnitude 10, direction 53.1° below +x
D magnitude 10, direction 53.1° above +x

115 Two perpendicular vectors have magnitudes 6 and 8. Their resultant is

A 14, at 53.1° from the 6-unit vector
B 10, at 53.1° from the 6-unit vector
C 10, at 36.9° from the 6-unit vector
D 2, at 53.1° from the 6-unit vector

116 If A = 2i - 3j and B = -4i + 5j, then A + B and its magnitude are

A 6i - 8j; 10
B -2i + 2j; 4
C -2i + 8j; 8.2
D -2i + 2j; 2.83

117 If A = i + 2j and B = 3i - j, then 2A - 3B =

A -7i + 7j
B 11i + j
C -7i + j
D 7i - 7j

118 A vector of magnitude 12 m points 40° north of east. Its components are

A east 7.71 m, north 9.19 m
B east 12 m, north 0
C east 9.19 m, north 7.71 m
D east 9.19 m, north 9.19 m

119 A hiker walks 5.0 km at 60° north of east, then 3.0 km east. The resultant displacement is

A 8.0 km at 60° north of east
B 7.0 km at 38° north of east
C 7.0 km at 52° north of east
D 6.1 km at 38° north of east

120 The vector (-4, 3) has magnitude and direction angle (counterclockwise from +x)

A 5; 143°
B 5; 37°
C 7; 143°
D 5; 217°

121 If A = (2, 3) and B = (-1, 4), then D = A - B and |D| are

A (1, 7); 7.07
B (3, 1); 3.16
C (-3, 1); 3.16
D (3, -1); 3.16

122 A vector of magnitude 8 at 150° from +x has components

A (6.93, 4.00)
B (-4.00, 6.93)
C (-6.93, 4.00)
D (-6.93, -4.00)

123 The vector that must be added to 3i + 2j to give zero is

A 3i + 2j
B -3i - 2j
C 2i + 3j
D -2i - 3j

124 Two vectors have magnitudes 3 and 5. The largest and smallest possible magnitudes of their sum are

A 8 and 2
B 15 and 2
C 8 and 0
D 4 and 2

125 A person walks 100 m east, 50 m north, then 30 m west. The resultant displacement is

A 180 m
B 86.0 m at 55° north of east
C 86.0 m at 35.5° east of north
D 86.0 m at 35.5° north of east

126 A vector of magnitude 10 has the same direction as A = 4i - 3j. It equals

A 4i - 3j
B 10i - 7.5j
C 8i - 6j
D 6i - 8j

127 A plane flies 200 km west then 150 km at 30° west of north. The resultant displacement is

A 304 km at 25.3° north of west
B 350 km at 30° north of west
C 304 km at 64.7° north of west
D 250 km at 25° north of west

128 A boat heads north at 4.0 m/s in a river whose current flows east at 3.0 m/s. Its velocity relative to the ground is

A 7.0 m/s
B 5.0 m/s at 36.9° east of north
C 5.0 m/s at 53.1° east of north
D 1.0 m/s north

129 Two forces, 10 N at 0° and 10 N at 120°, act on a body. The third force that makes the net force zero is

A 10 N at 120°
B 20 N at 60°
C 10 N at 60°
D 10 N at 240°

130 If A = 3i - 2j and B = -5i + 6j, the vector C with A + B + C = 0 and its magnitude are

A 2i + 4j; 4.47
B -2i - 4j; 4.47
C 2i - 4j; 4.47
D 8i - 8j; 11.3

131 The resultant of 50 N at 0°, 30 N at 90°, and 40 N at 225° is

A 120 N
B 21.8 N at 4.5° above +x
C 21.8 N at 85.5° above +x
D 50 N at 45°

132 A is 50 m at 20° north of east; B is 80 m at 40° south of east. A + B is

A 114 m at 17.6° south of east
B 130 m at 20° north of east
C 114 m at 17.6° north of east
D 74 m at 10° south of east

133 The vector 2i + aj (a > 0) has magnitude 5 when a =

A 3
B 5
C sqrt(21)
D 21

134 Two vectors of magnitudes 6 and 8 make a 60° angle. Their sum is

A 14
B 10, at 36.9° from the 6-unit vector
C 12.2, at 60° from the 6-unit vector
D 12.2, at 34.7° from the 6-unit vector

135 A particle moves from r1 = 2i + 3j (m) to r2 = -i + 5j (m). Its displacement is

A 3.61 m at 146° from +x
B 3.61 m at 34° from +x
C 5.83 m
D 1.0 m

136 For A = 2i - 3j + 6k, |A| and the unit vector are

A 11; (2i - 3j + 6k)/11
B 7; (2i - 3j + 6k)/7
C 7; 2i - 3j + 6k
D 5; (2i - 3j + 6k)/5

137 A vector of magnitude 12 points 25° above the negative x axis. Its components are

A (10.9, 5.07)
B (-5.07, 10.9)
C (-10.9, -5.07)
D (-10.9, 5.07)

138 If A = 2i + j, B = -3i + 4j, C = i - 6j, then A + B + C =

A 0
B i - j
C -j
D 3j

139 An orienteer runs 400 m at 30° north of east, then 600 m at 40° west of north. The total displacement is

A 661 m at 3.4° west of north
B 1000 m
C 661 m at 3.4° north of west
D 480 m at 30° north of east

140 To return to the start in Question 139, the orienteer must run

A 661 m at 3.4° west of south
B 661 m at 3.4° east of south
C 1000 m due south
D 661 m at 3.4° east of north

141 A vector of magnitude 26 pointing opposite to 5i + 12j is

A 10i + 24j
B 5i + 12j
C -5i - 12j
D -10i - 24j

142 Vector A has magnitude 10 at 30°. The vector B that makes A + B point along +y with magnitude 10 is

A 8.66i - 5.0j
B -8.66i - 5.0j
C -8.66i + 5.0j
D 8.66i + 5.0j

143 A plane with airspeed 200 km/h heads due north in a 50 km/h wind blowing toward the east. Its ground velocity is

A 206 km/h at 14.0° east of north
B 250 km/h at 14° east of north
C 206 km/h at 76° east of north
D 194 km/h at 14° east of north

144 To travel due north in that wind (airspeed 200 km/h), the pilot should head

A 14.5° east of north, ground speed 206 km/h
B due north, ground speed 150 km/h
C 14.5° west of north, ground speed 206 km/h
D 14.5° west of north, ground speed 194 km/h

145 For two vectors of magnitudes A and B with angle θ between them, the resultant magnitude R satisfies

A $R^2 = A^2 + B^2$
B $R^2 = A^2 + B^2 + 2AB\cos\theta$
C $R^2 = A^2 + B^2 - 2AB\cos\theta$
D $R = A + B$

146 Two vectors of magnitude 10 have a sum of magnitude 10. The angle between them is

A 60°
B 90°
C 120°
D 150°

147 For nonzero A and B, $|A + B| = |A - B|$ when the angle between them is

A 90°
B 0°
C 180°
D 45°

148 If $A = i + 2j$ and $B = 3i + j$, then $mA + nB = 7i + 4j$ for

A $m = 2, n = 1$
B $m = 1, n = 2$
C $m = 3, n = 1$
D $m = 1, n = 3$

149 A vector of magnitude 10 makes 40° with +x. In axes rotated 25° counterclockwise, its components (x', y') are

A (6.43, 7.66)
B (7.66, 6.43)
C (9.06, 4.23)
D (9.66, 2.59)

150 If $A = 2i - j + 3k$ and $B = -i + 4j + 2k$, then $|2A - B| =$

A 5.4
B 8.0
C 8.77
D 77

مفتاح الإجابات

1	C	2	A	3	D	4	B	5	A	6	D	7	C	8	B
9	D	10	A	11	B	12	C	13	A	14	D	15	B	16	C
17	A	18	D	19	B	20	C	21	A	22	B	23	D	24	C
25	A	26	D	27	B	28	A	29	C	30	D	31	B	32	C
33	A	34	D	35	B	36	A	37	C	38	B	39	D	40	A
41	C	42	B	43	D	44	A	45	C	46	B	47	D	48	A
49	C	50	B	51	B	52	D	53	A	54	C	55	B	56	A
57	D	58	C	59	A	60	B	61	D	62	C	63	B	64	A
65	C	66	D	67	B	68	A	69	C	70	D	71	A	72	B
73	D	74	C	75	A	76	B	77	C	78	D	79	B	80	A
81	D	82	C	83	A	84	B	85	D	86	C	87	B	88	A
89	C	90	D	91	A	92	C	93	B	94	D	95	B	96	A
97	C	98	D	99	A	100	B	101	C	102	A	103	D	104	B
105	C	106	A	107	D	108	B	109	A	110	C	111	B	112	D
113	A	114	C	115	B	116	D	117	A	118	C	119	B	120	A
121	D	122	C	123	B	124	A	125	D	126	C	127	A	128	B
129	D	130	C	131	B	132	A	133	C	134	D	135	A	136	B
137	D	138	C	139	A	140	B	141	D	142	C	143	A	144	D
145	B	146	C	147	A	148	B	149	D	150	C				